

WRLDZ / PROJECT ARGON

Spirit Lenses

*Custom binocular AR glasses from salvaged birdbath engines**Schematics · salvage hunt · bill of materials · assembly*

The Galaxy S23 Ultra is the computer. Do not gut it. The glasses are a stereo optical-see-through display plus IMU, built around sealed engines harvested from donor AR glasses (XREAL / Rokid / RayNeo / VITURE class).

Waveguides cannot be made in a garage. A phone taped to sunglasses is not this build. Alignment inside a birdbath engine **is** the product — keep engines sealed.

1. What you are building

Binocular optical-see-through AR glasses. Two birdbath engines (micro-OLED + 45° polarizing beam splitter + curved combiner), one USB-C DisplayPort pipe from the S23 Ultra. Black OLED pixels are off, so holograms float in the real room. Field of view is about 45–52° diagonal. The combiner throws away roughly 70–75% of world light (Karl Gutttag / Nreal Light teardown) — they look like sunglasses because of physics, not fashion.

Typical donor internals (XREAL Air 2): Sony ECX343E 1920×1200 per eye, Lontium LT7911UXC DP-to-dual-MIPI, MCU, IMU. Host supplies 5 V / ~1 A and video. Most of these glasses have no battery.

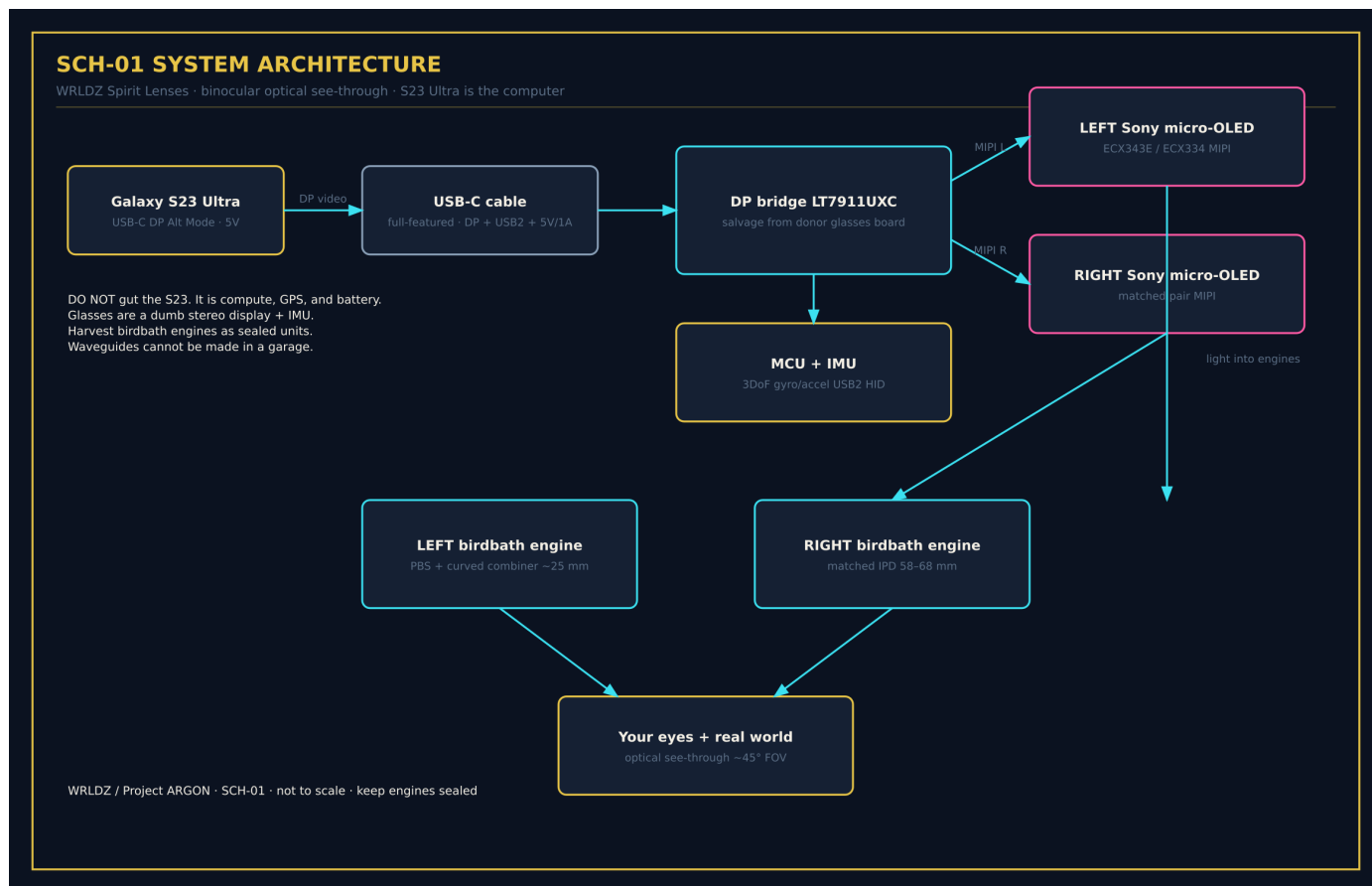
2. Order of work (do not skip)

- 1 Hunt one matched donor pair. Do not print a frame first.
- 2 First light in the donor frame with the S23. Prove both eyes and DisplayPort.
- 3 Caliper the engines. Schematic envelopes are typical, not gospel.
- 4 Print an open jig (no temples). Fuse a white rectangle in both eyes.
- 5 Print the frame, temples, strain relief. Wire. On-head first light.
- 6 IMU + Unity side-by-side. Cosmetics last.

3. Paths

Path	When	What you steal / buy
A Harvest	You found cracked XREAL / Rokid / RayNeo / VITURE	Sealed L+R engines + DP board + IMU + cable
B Modules	No living DP board	2× Sony ECX334/343 + drivers + LT7911-class bridge + birdbath kits
C Not this	Dead Quest 3	Video-passthrough helmet. Different product.

Schematic SCH-01 — system architecture



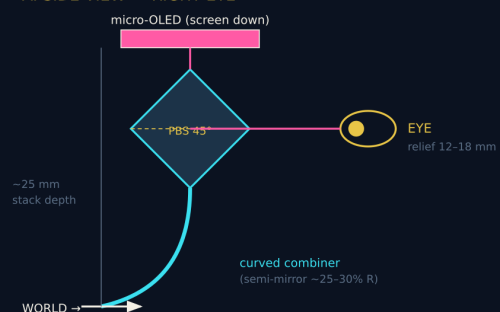
SCH-01 · S23 is compute. Glasses are display + IMU + optics.

Schematic SCH-02 — optical path

SCH-02 OPTICAL PATH (one eye, side) + BINOCULAR PLAN

Birdbath: OLED down into 45° polarizing beam splitter, folded onto a curved combiner. Depth ≈ 25 mm (Gutttag / Nreal Light).

A. SIDE VIEW — RIGHT EYE



Black OLED pixels = off = see-through. LCD donors stay milky. AMOLED / micro-OLED only.
Combiner blocks ~70-75% of world light (sunglasses). Do not walk in traffic while debugging FOV.

B. PLAN — BOTH EYES



FOV ≈ 45-52° diagonal (birdbath class)
Do not split a sealed engine. Alignment is the product.
Harvest LEFT and RIGHT as a matched pair from one donor.

WRLDZ / Project ARGON · SCH-02 · dimensions typical; measure YOUR donor engines before printing the frame

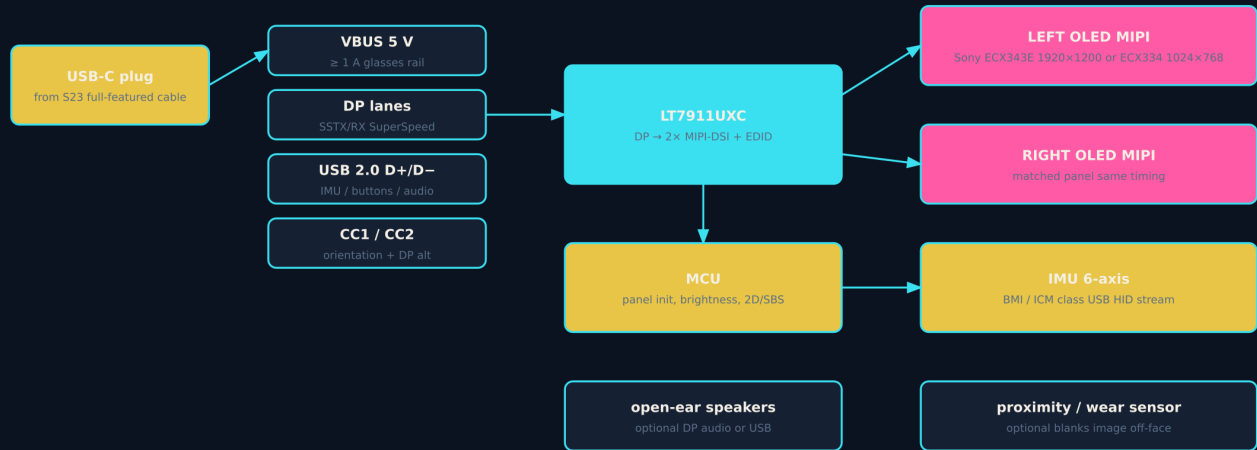
SCH-02 · One-eye birdbath (side) and binocular IPD plan. Stack depth ≈ 25 mm.

Schematic SCH-03 — electrical (functional)

SCH-03 ELECTRICAL (functional) USB-C DisplayPort glasses

S23 Ultra supplies 5 V / ~1 A + DP 1.2/1.4 video + USB 2.0 data. Glasses have no battery.

This is a FUNCTIONAL schematic of a harvested Xreal/Rokid/RayNeo-class board, not a PCB to etch.
If the donor DP board is alive: reuse it whole. If dead: buy a replacement LT7911UXC module or HDMI driver pair.



USB-C CABLE RULE

A charge-only cable will make the glasses look dead. Must carry SuperSpeed (DP) + 5 V.
Use the donor cable first. Then a marked "4K / 60 / video" USB-C cable. Angle plug toward the temple.
S23: Settings → Connected devices → Samsung DeX → Auto start when HDMI/DP connected = ON for first-light.
Unity later: render side-by-side (SBS) 3840×1080 (or 3840×1200). Long-press donor brightness+ 3 s for 3D/SBS if the MCU supports it.

IF THE DONOR BOARD IS DEAD

Path B: two HDMI micro-OLED driver boards (ECX334/335/343, 5 V, <2 W each) + USB-C DP to dual-HDMI splitter.
Stereo requires a splitter that can SBS-split or two cloned 2D images (no depth). Prefer a living LT7911 board.
Never power swollen cells. Donor glasses usually have NO pack — host 5 V only. Do not add a pack unless you know BMS.

WRLDZ / Project ARGON · SCH-03 · functional, not a PCB fab drawing

SCH-03 · Not a PCB to etch. Reuse a living donor board if you have one.

Schematic SCH-04 — mechanical

SCH-04 MECHANICAL FRAME + ENGINE POCKETS

Print the frame around MEASURED donor engines. Numbers below are envelopes — caliper your pair.

A. FRONT (looking at wearer)



B. ENGINE ENVELOPE (measure, then pad 0.4 mm)

Typical sealed birdbath engine	48 × 28 × 22 mm WxD×H (VERIFY)
Optical stack depth	~25 mm OLED → combiner
Eye relief	12-18 mm combiner to cornea
Temple USB-C	right or left angled cable strain-relief 15 mm
Nose bridge	print 3 heights 8 / 12 / 16 mm
Strap points	ski-goggle pins at temple rear 4 mm holes
Clearance to brows	≥ 6 mm foam strip

C. PRINT + FIT

1. Caliper both donor engines. They are often NOT identical left/right (cable exits differ).
2. Model pockets in CAD: engine + 0.4 mm slop, 1.5 mm walls, M2 heat-set inserts at 4 corners.
3. Print PETG or ABS (PLA creeps on skin heat), 0.16 mm layers on pockets.
4. First print: open jig (no temples) — confirm IPD and both eyes fuse a white rectangle.
5. Only then add temples, USB-C strain, foam, Millennium-Eye shell cosmetics.
6. Do not glue combiners. Engines drop in and screw down so you can service FPC cables.

WRLDZ / Project ARGON · SCH-04 · print AFTER donor engines are on the bench

SCH-04 · Print after the engines are on the bench. Caliper your pair.

4. Salvage hunt

Search copy-paste: *Xreal Air for parts · Nreal Light cracked · Rokid Max for parts · RayNeo Air 2 not working · Viture One cracked lens · AR glasses birdbath module*. Prefer cracked frame / dead USB-C with unscratched combiners. Sun-spotted combiners are trash — birdbaths focus sunlight onto the OLED.

Never: S23 Ultra, swollen cells, a phone panel ripped off its board, a sealed engine you have not proven dead.

Keep as a unit	Notes
Left + right birdbath engines	OLED+PBS+combiner sealed. Matched pair from one model.
DP board (LT7911UXC class) + FPC leads	Label L / R. Do not swap.
MCU + IMU	IMU often UV-glued to metal. Cut glue; do not pry MEMS.
Angled USB-C cable	Must carry SuperSpeed DP + 5 V. Charge-only cables fake a dead headset.
USB-C temple flex	Replace the receptacle if it wiggles.

Field test before you buy: photo of both combiners; USB-C wiggle; any image on a laptop DP port; same model left and right.

5. Bill of materials (one pair)

Qty	Part	Source
1	Matched L+R birdbath engines	Donor glasses
1	DP bridge + MCU + IMU	Same donor
1	Full-featured USB-C cable (4K60 video)	Donor or buy — not charge-only
1	Printed open jig, then frame (PETG/ABS, M2 inserts)	After calipers
8	M2×6 mm screws + brass heat-set inserts	Buy
2	EVA foam strips 8×80 mm	Brow / cheek light seal
1	Ski-goggle strap 20–25 mm + USB-C strain boot	Prototype retention
2*	Sony ECX334/343 + driver (Path B only)	Tindie / Taobao ECX kits
1*	LT7911UXC-class USB-C DP → dual MIPI/HDMI (Path B)	If donor board is dead

* Path B only. Two random HDMI boards clone 2D; they do not give stereo unless a single bridge SBS-splits.

Power: two OLEDs + bridge typically 0.7–1.5 W; MCU/IMU/speakers 0.2–0.5 W. Host VBUS 5 V / ~1 A. No extra pack in v1.

6. Assembly

Dark room. No sunlight on combiners. Tape glass edges. Do not stare at max-brightness OLED at 20 mm.

First light in the donor frame

S23: Settings → Connected devices → Samsung DeX → Auto start when HDMI connected = ON. Full-featured cable, angled plug into the temple. Both eyes should show DeX. Brightness+ hold ~3 s on many XREAL/Rokid units toggles 3D/SBS — leave 2D for this step. If black: another cable, a laptop DP port, the donor cable. Photograph through each combiner.

Teardown

Photo every screw. Temples off; support the USB-C flex. Rokid Max glues the outer lens — heat + alcohol, do not pry the combiner. Each engine: typically 3× Phillips plus a glue bead. Stop when it is a sealed brick with an FPC pigtail. Label L/R. Caliper W, D, H, FPC exit. Those six numbers override SCH-04.

Open jig, then frame

Pockets at 63 mm IPD, 0.4 mm slop, M2 inserts. White full-screen. You want one floating rectangle, not two. Slide IPD ± 2 mm, toe-in 1° if double. Do not live with a split image. USB-C on the same temple the donor used, 15 mm service loop, strain boot. IMU rigid to the engines, not a floppy temple tip. PETG, not PLA (creeps on skin heat). Strap for the prototype. Cosmetics last.

7. Bring-up (S23 + Unity)

Optics first. If the white rectangle does not fuse, software will not save it. DeX in both eyes = electrical path alive. Unity: black clear (holos only). 2D clone 1920×1080/1200 for first UI; SBS 3840×1080/1200 for depth. Pocket gyro is the wrong pose — the S23 is in a pocket. Use the glasses IMU (USB HID) or a BMI270 on the frame. Do not pipe phone-camera passthrough into birdbaths; the combiners are the world.

Pass: DeX both eyes; white field fused at 2 m; black field still shows the room; holos stay fused when you turn your head; walk without a split image.

6DoF later: SLAM cameras off an XREAL Air 2 Ultra, or keep ARCore on the S23 as a map companion only.

Rebuild drawings: python3 schematics/make_schematics.py then inkscape export. Sources: Gutttag Nreal Light birdbath (~25 mm stack, ~70–75% world-light loss); XREAL Air 2 teardown (ECX343E, LT7911UXC, 5 V host); S23 Ultra USB-C DisplayPort Alt Mode / DeX.